CS3003: IT Project Management

Dr Georgios Koutsoukos

giorgios.koutsoukos@rhul.ac.uk

CS3003: IT Project Management

Part 9: Fundamentals of Project Quality Management

Project Quality Management Fundamentals

Subject groups	Process groups				
	Initiating	Planning	Implementing	Controlling	Closing
Integration	4.3.2 Develop project charter	4.3.3 Develop project plans	4.3.4 Direct project work	4.3.5 Control project work 4.3.6 Control changes	4.3.7 Close project phase or project 4.3.8 Collect lessons learned
Stakeholder	4.3.9 Identify stakeholders		4.3.10 Manage stakeholders		
Scope		4.3.11 Define scope 4.3.12 Create work breakdown structure 4.3.13 Define activities		4.3.14 Control scope	
Resource	4.3.15 Establish project team	4.3.16 Estimate resources 4.3.17 Define project organization	4.3.18 Develop project team	4.3.19 Control resources 4.3.20 Manage project team	
Time		4.3.21 Sequence activities 4.3.22 Estimate activity durations 4.3.23 Develop schedule		4.3.24 Control schedule	
Cost		4.3.25 Estimate costs 4.3.26 Develop budget		4.3.27 Control costs	
Risk		4.3.28 Identify risks 4.3.29 Assess risks	4.3.30 Treat risks	4.3.31 Control risks	
Quality		4.3.32 Plan quality	4.3.33 Perform quality assurance	4.3.34 Perform quality control	
Procurement		4.3.35 Plan procurements	4.3.36 Select suppliers	4.3.37 Administer procurements	
Communication		4.3.38 Plan communications	4.3.39 Distribute information	4.3.40 Manage communications	

NOTE The purpose of this table is not to specify a chronological order for carrying out the activities. Its purpose is to map subject groups and process groups.

Project Quality Management Fundamentals

Definitions

- **Quality** (Prince2 and ISO 9000): "*The degree to which a set of inherent characteristics of a product, service, process, person, organization, system or resource fulfils requirements*."
- **Quality Management** (Prince2): "*The coordinated activities to direct and control an organization with regard to quality.*"
- Different Levels of Quality Management: **Organizational** Vs **Project**
  - At the **organizational (corporate)** level, quality management is concerned with **establishing and operating a framework** of organizational **processes** and **standards**, e.g. ISO 20000 (ITIL-based)
  - At the **project** level, quality management involves (i) the application of **specific processes** and verifying that these planned processes have been followed (typically according to the organizational processes) and (ii) ensuring that that the produced **products** meet established **quality criteria** (quality specifications and measurements)

Project Quality Management Fundamentals

Quality Management Process Groups (IS0 21500)

1. **Plan Quality** *(Planning)*
2. **Perform Quality Assurance** *(Implementing)*
3. **Perform Quality Control** *(Control Process group)*

Project Quality Management Fundamentals

Plan Quality (ISO 21500) *(Output: Quality Plan)*

"The purpose of Plan quality is to determine the quality requirements and standards that will be applicable to the project, the deliverables of the project and how the requirements and standards will be met based on the project objectives. This process includes the following:

- *determining and agreeing with the project sponsor and other stakeholders as to the objectives and relevant standards to be achieved;*
- *establishing the tools, procedures, techniques and resources necessary to achieve the relevant standards;*
- *determining methodologies, techniques and resources to implement the planned systematic quality activities;*
- *developing the quality plan which includes type of reviews, responsibilities and participants in a timetable in accordance with the project overall schedule;*
- *consolidating all quality information in the quality plan.*

Due to the temporary nature of projects and their time constraints, most projects do not have the ability to develop quality standards. Development and organizational acceptance of quality standards and product quality parameters may be outside of the project boundaries. This acceptance is normally the responsibility of the performing organization and serves as input to this process. The quality plan should refer to or include the quality policy as established by senior management."

Project Quality Management Fundamentals

Perform Quality Assurance (ISO 21500)

"*The purpose of* ***Perform quality assurance*** *is to review the deliverables and the project. It includes all processes, tools, procedures, techniques and resources necessary to meet quality requirements.*
This process includes the following:

- *ensuring objectives and relevant standards to be achieved are communicated, understood, accepted and adhered to by the appropriate project organization members;*
- *executing the quality plan as the project progresses;*
- *ensuring that the established tools, procedures, techniques and resources are being used.*

Quality assurance audits ***may be performed outside the project boundaries by other parts of the performing organization or by the customers****.* ***Audits*** *determine the performance of the quality* ***process****, quality* ***control*** *and the* ***need for recommended action or change requests."***

In essence, "*Quality assurance provides a check that the project's direction and management are adequate for the nature of the project and that it complies with relevant corporate or programme management standards and policies*." (Prince2)

Project Quality Management Fundamentals

Quality Assurance Vs Project Assurance (Prince2)

*"**Quality** **assurance** provides a check that the project's direction and management are adequate for the nature of the project and that it complies with relevant **corporate, programme management or customer standards** and **policies**. Quality assurance is therefore **independent** of the project."*

*"**Project** **assurance** is the **project board's responsibility** to **assure itself** that the **project is being conducted correctly**. The project board members each have a specific area of focus for project assurance, namely business assurance for the executive, user assurance for the senior user(s) and supplier assurance for the senior supplier(s). Project assurance is therefore **independent of the project manager** but **not independent of the project."***

Project Quality Management Fundamentals

Perform Quality Control (ISO 21500)

"*The purpose of* ***Perform quality control*** *is to determine whether the established project objectives, quality requirements and standards are being met and to identify causes of, and ways to eliminate, unsatisfactory performance. This process should be applied during the whole project life cycle and includes the following:*

- *monitoring the quality of the deliverables and processes is being met and detecting defects by using the established tools, procedures and techniques;*
- *analysing possible causes of defects;*
- *determining the preventive actions and change requests;*
- *communicating the corrective actions and change requests to the appropriate project organization members.*

Quality control may identify causes of poor process performance or product quality and may result in recommended actions or change requests, when necessary to eliminate non-conforming performance.

In essence (Prince2), "*Quality control focuses on the operational techniques and activities used by those involved in the project to:*

- *Fulfil the requirements for quality (for example, by quality inspections or testing)*
- *Identify ways of eliminating causes of unsatisfactory performance (for example, by introducing process improvements as a result of lessons learned)."*

Project Quality Management Fundamentals

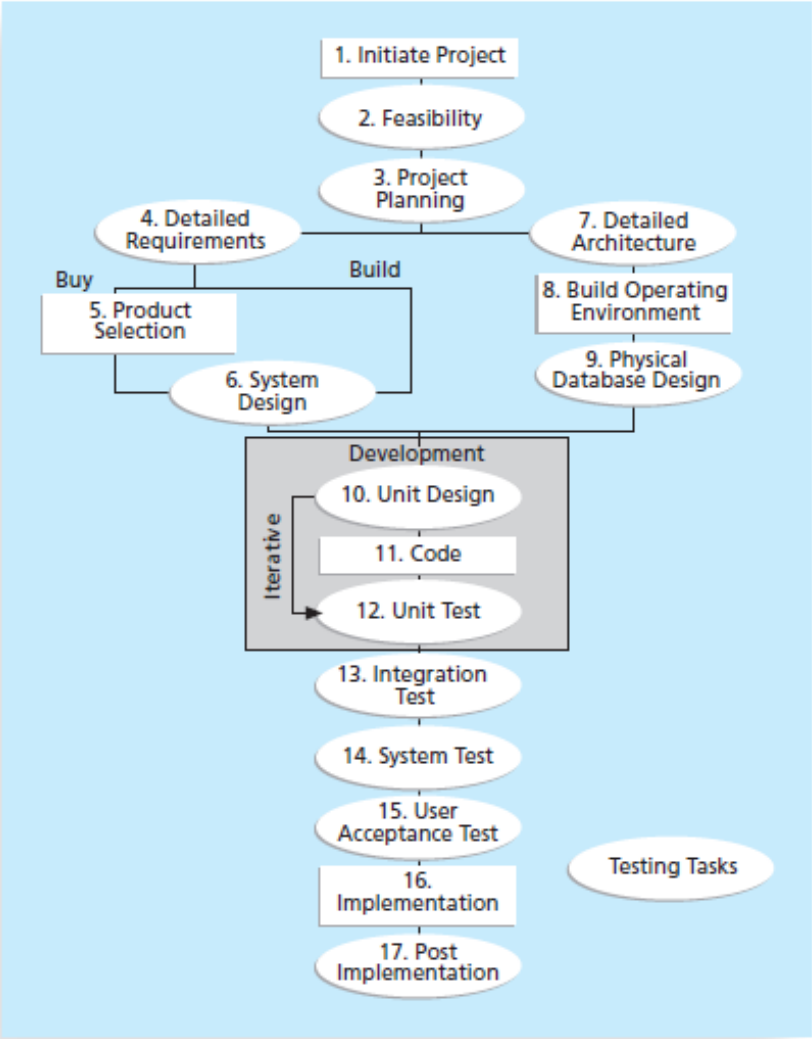
Quality Register (Prince2)

"Summarize all the quality management activities that are planned or have taken place, and provides information for the end stage reports and end project report."

Quality Activity ID	Product ID	Product	Quality Method	Producer	Reviewer(s)	Approver(s)	Target Review Date	Actual Review Date	Target Approval Date	Actual Approval Date	Result

Project Quality Management Fundamentals

Software/IT Systems Testing Overview



Source: Hollstadt & Associates, Inc.

Image from Schwalbe's book .

+ Performance/Stress Testing
+ Security Testing
+ Scalability and High-Availability Testing
+ Disaster Recovery Testing
+….

Note that there are often difference in terminology to refer to similar "things" with "some" differences (e.g. Integration Vs System Vs Regression Vs Release Testing).

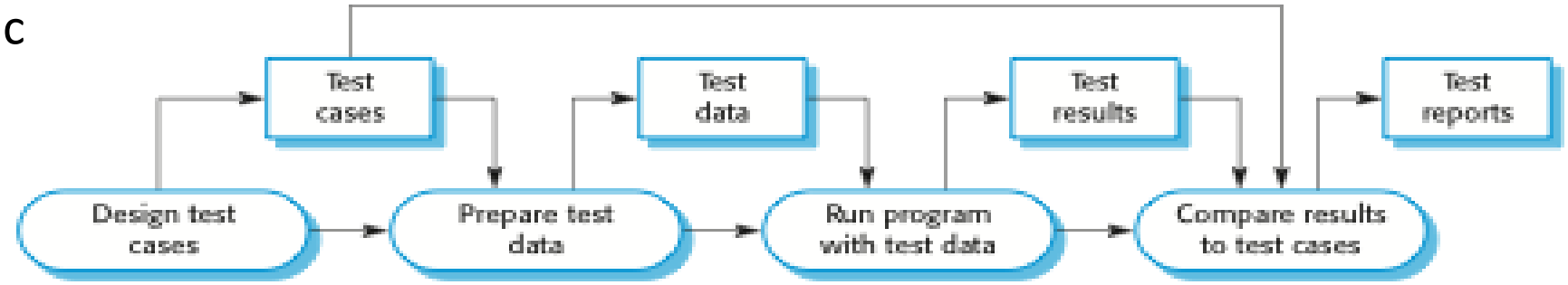
Who performs the testing is also a key factor…

Manual (typically tool assisted) vs Automated?

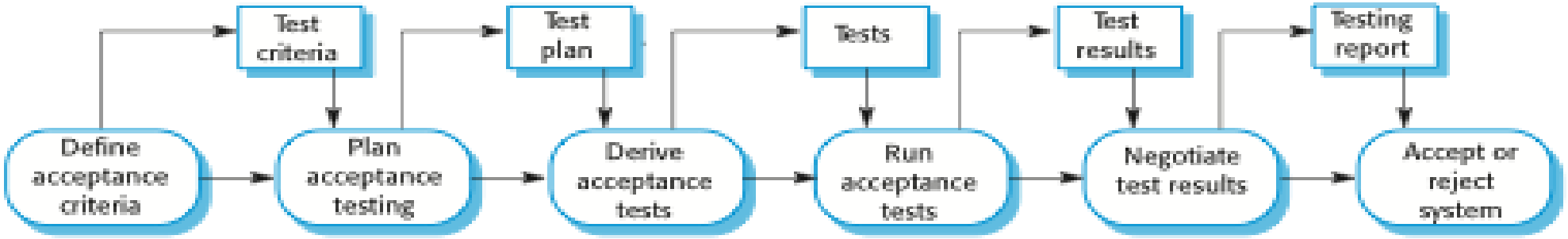
Project Quality Management Fundamentals

Software/IT Systems Testing Process

Generic



UAT



Project Quality Management Fundamentals

Common Software Quality Attributes

- There is often a tension between customer/end user quality requirements (efficiency, reliability, etc.) and developers quality requirements (maintainability, reusability, etc.)
- Some quality requirements are difficult to specify in an unambiguous way
- It is very difficult for any system to be optimized for all of these attributes. Therefore, the quality plan should define the most important quality attributes. The plan should also include a definition of the quality assessment process, an agreed way of assessing/measuring whether desired quality attributes are present in the product.

Safety	**Understandability**	**Portability**
Security	**Testability**	**Usability**
Reliability	**Adaptability**	**Reusability**
Resilience	**Modularity**	**Efficiency**
Robustness	**Complexity**	**Learnability**

Project Quality Management Fundamentals

Quality Standards

- Definition (ISO): *"requirements, specifications, guidelines or characteristics that can be used consistently to ensure that materials, products, processes and services are fit for their purpose*." Guidance Vs Requirements standards.
- Advantages of standards: encapsulation of best practice- avoids repetition of past mistakes, framework for defining what quality means in a particular setting i.e. that organization's view of quality, they provide continuity - new staff can understand the organisation by understanding the standards that are used.
- **ISO 9000** (principles) and **ISO 9001** (requirements): General standards that can be used as a basis for developing quality management systems.

  Can be considered as a "framework" for developing organizational quality standards. It sets out general quality principles, describes quality processes in general and lays out the organizational standards and procedures that should be defined. These should be documented in an organizational quality manual.

  In essence, ISO 9001 governs the implementation and maintenance of a quality management system, ensures organizations' processes are managed, documented, and continually improve. It is applicable to any organization and industry (and may be "certified").

Project Quality Management Fundamentals

ISO 9001 Quality Standard

Project Quality Management Fundamentals

Continuous Process Improvement Process

- A crucial requirement/characteristic/attribute. Part of overall Total Quality Management (TQM).
- There is no ‘ideal’ or ‘standard’ process that is applicable in all organizations or for all software/IT products/projects of a particular type. It is rarely successful in introducing process improvements if you simply attempt to change the process to one that is used elsewhere. One should consider the local environment and culture and how this may be affected by process change proposals.
- Each company typically develops its own process depending on its size, the background and skills of its staff, the type of software/IT system being developed, customer and market requirements, and the company culture.
- In continuous process improvement one has to consider what aspects of the process to improve (e.g. product quality, process itself (e.g. development time)?

Project Quality Management Fundamentals

Continuous Process Improvement Process

There are several approaches in Process Improvement. Common Steps:

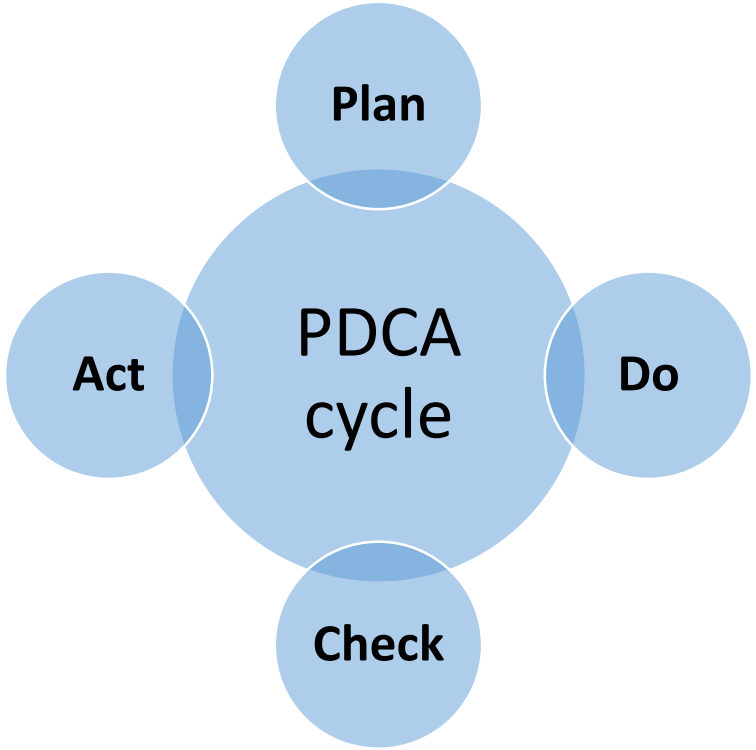
1. **Identification of goals**
2. **Analysis of the present situation**
3. **Development of an approach/solution**
4. **Construction of a plan**
5. **Execution of the plan**
6. **Measurement of results**

Project Quality Management Fundamentals

Continuous Process Improvement: Plan–Do–Check–Act (PDCA) cycle

Act

Determine next steps: If the desired goal was not achieved, repeat the PDCA process. If the goal was achieved, identify systemic changes needed for full implementation. Adopt the solution and monitor results. Look for the next improvement opportunity.



Plan

Identify the problem (select problems to be analyzed and establish a precise problem statement. Set measurable goals for the problem solving effort.
Analyze the problem: Identify the processes that impact the problem and select one; List the steps in the process as it currently exists ; Identify potential cause of the problem; Collect and analyze data related to the problem; Verify or revise the original problem statement; Identify root causes of the problem.

Do

Establish criteria for selecting a solution. Generate potential solutions that address the root causes of the problem. Select a solution. Plan the solution implementation. Implement a solution: Implement the chosen solution on a trial or pilot basis.

Check

Evaluate the results: Gather data on the solution. Analyze the data on the solution.

Descriptions from the Boris Mutafelija and Harvey Stromberg Book on Continuous Process Improvement